

PRODUCT SPECIFICATION

WC-100 INDUSTRIAL WEAR COMPONENT

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DOCUMENT CONTROL

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Revision	A
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Primary Audience	Engineering, Manufacturing, Quality and Maintenance
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Classification	Portfolio Demonstration
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1. PURPOSE

This specification defines the technical, dimensional, material, manufacturing and quality requirements for the WC-100 Industrial Wear Component.

The purpose of this document is to establish a consistent reference for engineering, manufacturing, inspection, quality and maintenance activities associated with the product.

The specification provides controlled requirements that can be used to verify whether a manufactured component conforms to the defined product criteria.

2. SCOPE

This specification applies to the manufacture, inspection, identification and acceptance of the WC-100 Industrial Wear Component.

It covers:

- Product description
- Material requirements
- Dimensional requirements
- Manufacturing requirements
- Surface and finishing requirements
- Quality and inspection requirements
- Identification and traceability
- Packaging requirements
- Acceptance criteria
- Documentation and revision control

This document does not define detailed manufacturing process parameters for individual production machines. Such information should be maintained in the applicable manufacturing work instructions and process documentation.

3. PRODUCT DESCRIPTION

The WC-100 is a fictional industrial wear component designed for use in equipment operating in demanding material-handling and heavy-industrial environments.

The component is intended to provide a replaceable wear surface in applications where repeated contact with abrasive material may result in component degradation.

The design emphasizes:

- Wear resistance

- Dimensional consistency
- Structural integrity
- Maintainability
- Traceability
- Repeatable manufacturing quality

The WC-100 may be supplied as an individual component or as part of a larger equipment assembly.

4. INTENDED APPLICATION

The WC-100 is intended for industrial applications involving repeated mechanical contact and exposure to abrasive operating conditions.

Typical application environments may include:

- Material handling equipment
- Processing equipment
- Industrial conveying systems
- Mining-related equipment
- Heavy manufacturing equipment

Final suitability for a specific application must be confirmed by the responsible engineering function based on operating conditions, loading, material characteristics and equipment design.

5. MATERIAL REQUIREMENTS

The WC-100 shall be manufactured using an approved wear-resistant steel grade suitable for the intended application.

5.1 Material Requirements

Property	Requirement
Base Material	Wear-resistant alloy steel
Material Certification	Required
Material Traceability	Required

Heat/Batch Identification	Required
Surface Condition	Free from unacceptable cracks, laminations and significant surface defects
Material Substitution	Requires engineering approval

The selected material grade shall comply with the applicable engineering specification and approved material certificate.

Material certificates shall be retained according to the applicable document retention and quality requirements.

6. DIMENSIONAL REQUIREMENTS

The following dimensions define the baseline configuration of the WC-100 demonstration product.

Dimension	Nominal Value	Acceptance Requirement
Overall Length	250 mm	±1.0 mm
Overall Width	150 mm	±1.0 mm
Overall Thickness	25 mm	±0.5 mm
Mounting Hole Diameter	22 mm	±0.2 mm
Hole Centre Distance	100 mm	±0.5 mm
Edge Radius	5 mm	±1.0 mm

All dimensions shall be verified using calibrated inspection equipment appropriate to the measurement requirement.

Any dimensional characteristic identified as critical by engineering shall be subject to the applicable additional inspection requirement.

7. GEOMETRIC REQUIREMENTS

The component shall maintain dimensional consistency sufficient to ensure proper installation and intended operation.

The following baseline requirements apply:

- Major surfaces shall be reasonably flat and free from unacceptable distortion.
- Mounting holes shall be positioned within the specified centre-distance tolerance.
- Edges shall be free from excessive burrs and sharp projections.
- Component geometry shall remain within the approved drawing or product definition.
- Any deformation affecting fit, installation or intended function shall be treated as a non-conformance.

Where a controlled engineering drawing exists, the drawing shall take precedence for product-specific geometric requirements.

8. MANUFACTURING REQUIREMENTS

The WC-100 shall be manufactured using approved production processes and equipment operated by appropriately trained personnel.

Manufacturing activities may include:

1. Material identification and verification
2. Material preparation
3. Cutting or forming
4. Machining or drilling
5. Edge finishing
6. Cleaning
7. Inspection
8. Identification and traceability
9. Final quality verification
10. Packaging

Detailed production parameters shall be defined in controlled manufacturing process documentation and work instructions.

Changes to an approved manufacturing process shall be evaluated through the applicable engineering and change-management process.

9. SURFACE AND FINISH REQUIREMENTS

The finished component shall have a surface condition appropriate for its intended application.

The component shall be free from:

- Visible cracks
- Significant laminations
- Unacceptable deformation
- Excessive burrs
- Loose material
- Contamination that could affect product quality
- Surface damage that could affect fit or function

Minor cosmetic variations that do not affect product performance may be accepted in accordance with applicable quality criteria.

10. QUALITY AND INSPECTION REQUIREMENTS

Inspection activities shall verify that the finished WC-100 meets the specified product requirements before release.

10.1 Incoming Material Inspection

Material shall be verified against:

- Material identification
- Material certificate
- Heat or batch number
- Approved material requirement

10.2 In-Process Inspection

Manufacturing personnel or quality personnel shall verify applicable dimensions and process characteristics during production.

10.3 Final Inspection

Final inspection shall verify, as applicable:

- Overall dimensions
- Thickness
- Hole diameter
- Hole location
- Surface condition
- Product identification
- Traceability
- Visual condition

Inspection results shall be recorded using the applicable quality record or inspection documentation.

11. MEASUREMENT EQUIPMENT

Inspection equipment shall be suitable for the required measurement accuracy and maintained according to the applicable calibration program.

Examples of measurement equipment may include:

- Vernier caliper
- Micrometer
- Steel rule
- Bore gauge
- Height gauge
- Measuring tape
- Appropriate dimensional inspection equipment

Where calibrated equipment is required, only equipment within its valid calibration period shall be used.

12. PRODUCT IDENTIFICATION AND TRACEABILITY

Each WC-100 component shall be identifiable through an appropriate product or batch identification method.

Traceability information may include:

- Product number
- Part number
- Revision
- Manufacturing batch
- Heat number
- Production date
- Inspection status

Identification shall remain legible throughout the applicable product handling and storage stages.

Traceability information shall be linked to the applicable manufacturing and quality records where required.

13. ACCEPTANCE CRITERIA

The WC-100 shall be accepted for release when:

1. Required material documentation is available.
2. Specified dimensions meet the applicable tolerances.
3. Required inspection activities have been completed.
4. No unacceptable surface or manufacturing defects are identified.
5. Product identification and traceability requirements are satisfied.
6. Required quality records have been completed.
7. Any identified non-conformances have been appropriately dispositioned.
8. The product has received the required release approval.

Products that do not meet specified requirements shall be controlled through the applicable non-conformance process.

14. NON-CONFORMANCE MANAGEMENT

Any component that fails to meet a specified requirement shall be identified and controlled to prevent unintended use.

Potential dispositions may include:

- Rework
- Repair
- Use-as-is with appropriate approval
- Scrap
- Engineering review

The disposition of a non-conforming product shall be documented according to the applicable quality-management process.

Engineering approval shall be obtained where a deviation from a defined product requirement could affect fit, function, safety or performance.

15. PACKAGING AND IDENTIFICATION

The WC-100 shall be packaged in a manner that protects the component from damage during handling, storage and transportation.

Packaging shall prevent, as applicable:

- Excessive surface damage
- Contamination
- Loss of product identification
- Uncontrolled movement during transportation

The package or shipping documentation shall include sufficient identification to maintain product traceability.

16. STORAGE AND HANDLING

Components shall be stored in conditions appropriate for the material and intended application.

During handling:

- Components shall not be dropped or handled in a manner that could cause damage.

- Product identification shall be maintained.
 - Components shall be protected from conditions that could adversely affect product quality.
 - Damaged components shall be segregated and evaluated before use.
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17. DOCUMENTATION REQUIREMENTS

The following documentation may be associated with the WC-100 product lifecycle:

Document	Purpose
Product Specification	Defines product requirements
Engineering Drawing	Defines controlled product geometry
Manufacturing Work Instruction	Defines production activities
Inspection Plan	Defines inspection requirements
Quality Record	Records inspection results
Material Certificate	Provides material verification
Technical Manual	Provides product-related technical information
User Guide	Supports correct product use
Release Notes	Communicates controlled product changes
Knowledge Base Article	Provides troubleshooting/support information

All controlled documents shall be maintained according to the applicable document-management and lifecycle requirements.

18. DOCUMENT LIFECYCLE

Controlled documentation associated with the WC-100 should follow a defined lifecycle:

Plan → Draft → Technical Review → SME Review → Approval → Release → Publish → Maintain → Revise → Supersede

Document revisions shall be traceable to an approved change or update.

Superseded documents shall be appropriately identified and prevented from unintended use where required.

19. CHANGE MANAGEMENT

Changes affecting product requirements, manufacturing processes or supporting documentation shall be evaluated before implementation.

A documentation impact assessment should consider:

- Product specification
- Engineering drawings
- Manufacturing work instructions
- Inspection documentation
- Technical manuals
- User guides
- Training materials
- Knowledge articles
- Release notes

The responsible stakeholders shall determine which documents require revision as a result of the change.

20. EXAMPLE ENGINEERING CHANGE

Change Request

Change: Increase mounting hole diameter from 20 mm to 22 mm.

Documentation Impact

Document	Impact	Required Action
Product Specification	High	Revise

Engineering Drawing	High	Revise
Manufacturing Work Instruction	High	Revise
Inspection Plan	High	Revise
Technical Manual	Medium	Review
User Guide	Medium	Review
Training Guide	Medium	Review
Release Notes	Required	Create
Knowledge Base Article	Review	Assess impact

This example demonstrates how a technical writer can assess downstream documentation impacts when a product or engineering requirement changes.

21. DOCUMENT MANAGEMENT ENVIRONMENT

Technical documentation may be maintained within controlled document-management or product-lifecycle environments.

Relevant systems may include:

Teamcenter

Used conceptually for managing product-related information, revisions, datasets, product structures, workflows and controlled lifecycle states.

Windchill

Used conceptually for managing product documentation, revisions, lifecycle states, workflows and engineering change processes.

SharePoint

Used for enterprise document repositories, collaboration, controlled content, version management and knowledge sharing.

The appropriate system of record shall be determined by the organisation's established processes and governance requirements.

22. ROLES AND RESPONSIBILITIES

Technical Writer

- Gather technical information from relevant stakeholders.
- Develop and maintain documentation.
- Structure information for the intended audience.
- Coordinate technical and SME reviews.
- Maintain document revisions.
- Incorporate approved changes.
- Ensure documentation is clear, accurate and consistent.
- Support documentation lifecycle activities.

Subject Matter Expert

- Provide technical information.
- Validate technical accuracy.
- Review proposed documentation changes.
- Clarify technical requirements.

Engineering

- Define product and engineering requirements.
- Approve applicable technical changes.
- Support engineering change activities.

Quality

- Define and verify applicable quality requirements.
- Review inspection and acceptance criteria.
- Support non-conformance management.

Manufacturing

- Provide process information.
- Validate manufacturing documentation.
- Execute approved manufacturing processes.

23. RELATED DOCUMENTATION

The following supporting documents may be associated with this specification:

- WC-100 Engineering Drawing
- WC-100 Manufacturing Work Instruction
- WC-100 Inspection Plan
- WC-100 Technical Manual
- WC-100 User Guide
- WC-100 Operator Training Guide
- WC-100 Release Notes
- WC-100 Knowledge Base Article

These documents would be maintained under the applicable document-management and revision-control process.

24. REVISION HISTORY

Revision	Date	Description	Author
A	30-Aug-2026	Initial demonstration release	Technical Documentation

25. PORTFOLIO DEMONSTRATION NOTICE

This document is an independently created technical writing demonstration developed for a professional portfolio.

WC-100 Industrial Wear Component is a fictional product.

All product characteristics, dimensions, technical requirements, manufacturing scenarios and specifications contained in this document are fictional and have been created solely to

demonstrate technical writing, structured documentation, manufacturing documentation concepts and document-control practices.

This document does not contain confidential, proprietary or customer information belonging to any employer or third party.

END OF DOCUMENT

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